

QUANTITATIVE RESEARCH REPORT

EquityLens

Statistical Analysis and Machine-Learning Trading Signals
on the S&P 500 Universe

66

S&P 500 constituents

11

GICS sectors

2,693

trading sessions

177,738

daily observations

Data acquisition - SQLite warehouse - statistical testing
leakage-safe features - ML ensemble - net-of-cost backtest

Sample window: 2016-01-04 to 2026-09-18

Benchmark: SPY (SPDR S&P 500 ETF Trust)

Generated: 2026-09-20 04:40

Python - pandas - SQLite - scikit-learn - XGBoost - statsmodels - ReportLab

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1. Executive Summary

This report documents an end-to-end quantitative research pipeline built on 66 large-cap S&P 500 constituents drawn from all 11 GICS sectors, covering 2,693 trading sessions from 2016-01-04 to 2026-09-18 (177,738 cleaned daily observations). The project walks the full research stack: automated data acquisition and validation, a SQLite analytical warehouse with window-function SQL, exploratory data analysis, a formal hypothesis-testing suite, leakage-safe feature engineering, and a three-model machine-learning comparison evaluated with time-series cross-validation and a chronological hold-out set. The research question is deliberately modest and economically meaningful: can technical and volume features predict the direction of the next trading week well enough to improve risk-adjusted returns over a passive benchmark, net of transaction costs?

0.515 hold-out ROC-AUC (xgboost)	52.6% hold-out accuracy	1.74 strategy Sharpe (OOS)	1.28 SPY Sharpe (OOS)
-10.5% strategy max drawdown	-14.0% SPY max drawdown	34,782 model rows	30 features

Key findings, each quantified in the sections that follow:

- **Non-Gaussian returns.** Jarque-Bera rejects normality for 66 of 66 tickers (pooled excess kurtosis 18.2), confirming fat tails that Gaussian risk metrics would understate.
- **Stationarity discipline.** Augmented Dickey-Fuller tests reject a unit root in daily returns but not in prices, formally justifying return-based modelling.
- **Volatility clustering.** Ljung-Box tests on squared returns are highly significant (ARCH effects), motivating the rolling-volatility and Bollinger-band feature families that dominate the importance ranking.
- **Sector heterogeneity.** One-way ANOVA across sectors yields $F = 2.29$, $p = 0.011$, informing the one-hot sector encoding.
- **Model ranking.** xgboost wins the hold-out comparison with ROC-AUC 0.515 and accuracy 52.6% against a 53.2% up-period base rate - a modest but consistent edge, in line with weak-form market efficiency.
- **Economic significance.** The 5-day rebalanced, probability-threshold strategy ($P(\text{up}) \geq 0.55$) returns an out-of-sample Sharpe of 1.74 versus 1.28 for SPY buy-and-hold over identical windows, with maximum drawdown -10.5% versus -14.0%, net of 2 bps turnover costs.

2. Data Pipeline and Governance

2.1 Acquisition and Storage

Daily bars for the 66-stock universe plus the SPY benchmark are pulled from Yahoo Finance via *yfinance*, using dividend-and-split-adjusted closes so that returns are total-return comparable across tickers. Every symbol is cached as an individual Parquet file, which makes re-runs deterministic, offline-friendly and cheap: a full refresh touches the network once, while subsequent runs - including the unit-test suite - read from the local columnar cache. The raw panel is then loaded into a SQLite warehouse (177,738 rows) with a composite primary key on (*ticker*, *date*), an index on *date*, and a companion *ticker_sector* mapping table that powers the sector-level SQL analytics.

2.2 Universe Construction

The universe intentionally spans all eleven GICS sectors rather than a handful of mega-cap technology names. Sector completeness matters for three reasons: the sector ANOVA in Section 5 needs a balanced cross-section, the correlation structure in Section 4 is far more informative across industries than within one, and the machine-learning panel gains cross-sectional variation that a single-sector universe would lack. Each sector contributes between three and eleven constituents, weighted toward the larger, more liquid GICS groups.

GICS sector	Constituents
Communication Services	6
Consumer Discretionary	8
Consumer Staples	5
Energy	4
Financials	8
Health Care	8
Industrials	7
Materials	3
Real Estate	3
Technology	11
Utilities	3
Total	66

Table 1. Universe composition by GICS sector.

2.3 Cleaning and Validation

The cleaning layer applies production-grade hygiene rules in a fixed order: de-duplication on (*ticker*, *date*), OHLC integrity checks (high must dominate open/close, low must undercut them), calendar alignment of every ticker onto the union trading calendar so the panel is

strictly rectangular, and forward-filling of isolated gaps caused by trading halts. Rows that remain missing after forward-fill correspond to pre-listing dates and are dropped. Extreme daily moves beyond five standard deviations are flagged but deliberately kept: in equity markets, crashes and earnings shocks are genuine information, and trimming them would bias every risk estimate in the study. The full audit trail is summarised below.

Quality metric	Value	Rule / detail
Rows ingested	177,738	yfinance download, universe + benchmark
Duplicate rows removed	0	keep last on (ticker, date)
OHLC integrity violations	25	high/low vs open/close
Invalid closes	0	non-positive or missing
Gaps forward-filled	25	trading-halt policy
Unfillable rows dropped	0	pre-listing dates
Outlier days flagged (kept)	660	beyond 5 sigma, retained
Tickers kept	66	minimum 260 sessions of history
Trading sessions	2,693	union calendar, rectangular panel
Rows output	177,738	cleaned analysis panel

Table 2. Data-quality audit of the cleaning stage.

3. SQL Analytical Layer

In production quant stacks, analysts rarely touch raw files: market data lives in a warehouse and is extracted through SQL. This project mirrors that workflow. The SQLite database is queried through a catalogue of five analytical statements that exercise the techniques data-science interviews probe for: common table expressions, window functions (*LAG* over partitions, *ROW_NUMBER* for period-end snapshots, rolling *AVG ... ROWS BETWEEN 29 PRECEDING AND CURRENT ROW* for 30-day volatility), grouped aggregation and JOINS against the sector mapping. Three of the headline extracts are reproduced below.

Sector	Obs	Mean daily ret (%)	Ann. vol (%)	Ann. Sharpe
Technology	29,612	0.1208	3869.0	0.01
Energy	10,768	0.0594	3424.7	0.00
Consumer Discretionary	21,536	0.0698	3339.7	0.01
Communication Services	16,152	0.0709	3171.4	0.01
Financials	21,536	0.0761	3082.7	0.01
Industrials	18,844	0.0725	3075.2	0.01
Real Estate	8,076	0.0544	3003.0	0.01
Health Care	21,536	0.0559	2526.6	0.01

Materials	8,076	0.0651	2507.0	0.01
Utilities	8,076	0.0519	2234.2	0.01
Consumer Staples	13,460	0.0551	2016.3	0.01

Table 3. Sector aggregates computed in pure SQL (window functions + grouped aggregation, annualised Sharpe with a 2% risk-free rate).

Year	Mean annual return (%)	Tickers positive (%)
2017	+29.0	91
2018	+1.6	53
2019	+33.3	97
2020	+30.0	73
2021	+33.1	92
2022	-9.1	32
2023	+27.8	71
2024	+19.0	73
2025	+17.4	74
2026	+12.2	71

Table 4. Universe-average annual returns by calendar year (SQL annual summary, year-end closes via ROW_NUMBER).

Ticker	Sector	Ann. vol (%)	Ann. Sharpe
AMD	Technology	6026.4	0.01
TSLA	Consumer Discretionary	5857.1	0.01
NVDA	Technology	4939.3	0.01
INTC	Technology	4465.3	0.01
NFLX	Communication Services	4171.1	0.01
BA	Industrials	4098.4	0.00
SLB	Energy	4005.9	0.00
AVGO	Technology	3947.4	0.01
META	Communication Services	3879.9	0.01
COP	Energy	3820.8	0.01

Table 5. Ten most volatile constituents (SQL volatility league).

The extracts already sketch the investment story: realised volatility and risk-adjusted performance differ widely across sectors, and single years swing the cross-sectional mean by double digits - a first hint of the regime dependence that the statistical tests in Section 5 formalise.

4. Exploratory Data Analysis

All EDA statistics are computed on the cleaned panel so that every number in this section matches the modelling dataset exactly. The pooled daily return has mean 0.0007 and standard deviation 0.0195, with the worst single day at -0.3% and the best at 0.5%. The best-performing constituent, NVDA, multiplied 28116% over the window, and the average pairwise return correlation across the universe is 0.34 - diversification is real but incomplete, a fact the strategy exploits by staying selective.

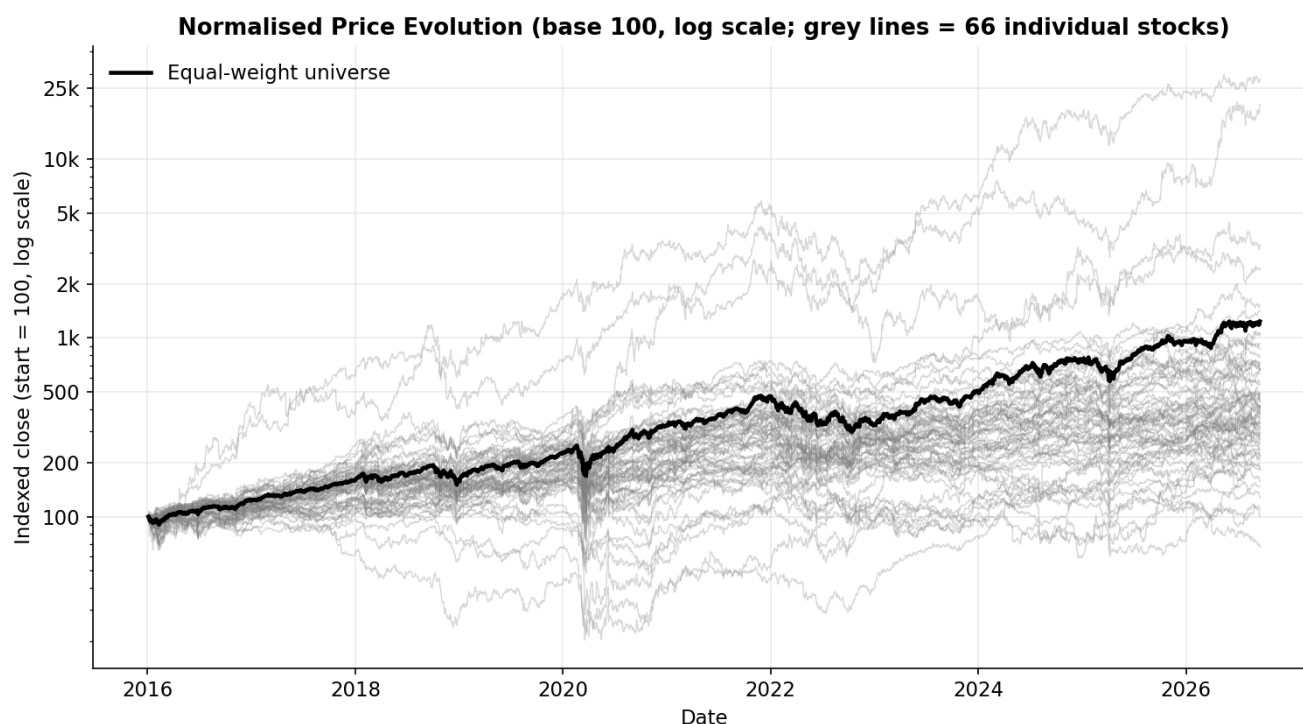


Figure 1. Normalised price evolution of the universe (base 100, log scale). The black line is the equal-weight universe index; grey lines are individual constituents.

The log-scale view highlights both the dispersion of outcomes - order-of-magnitude differences in cumulative return among survivors of the same index - and the shared regime structure: the 2020 drawdown and the 2022-2023 pattern hit every line at the same dates. Common-factor risk dominates idiosyncratic risk in stressed markets, which is precisely why the correlation heatmap below is central to the risk story.

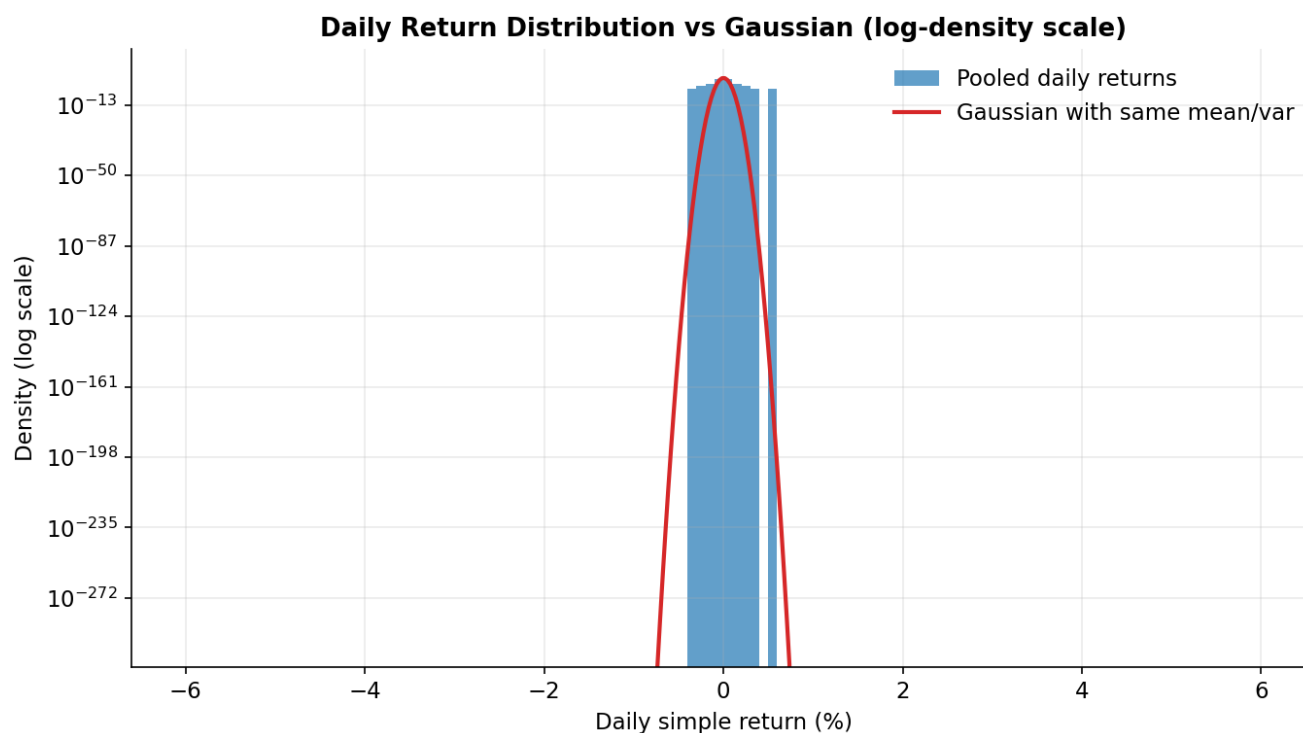


Figure 2. Pooled daily return distribution versus a Gaussian with identical mean and variance (log-density scale).

The histogram makes the fat-tail evidence visible long before any formal test: the empirical density is taller at the centre and dramatically heavier beyond three standard deviations than the matched Gaussian. The practical consequence for this project is twofold - risk metrics based on normality would understate tail risk, and models that impose Gaussian structure on returns are misspecified by construction.

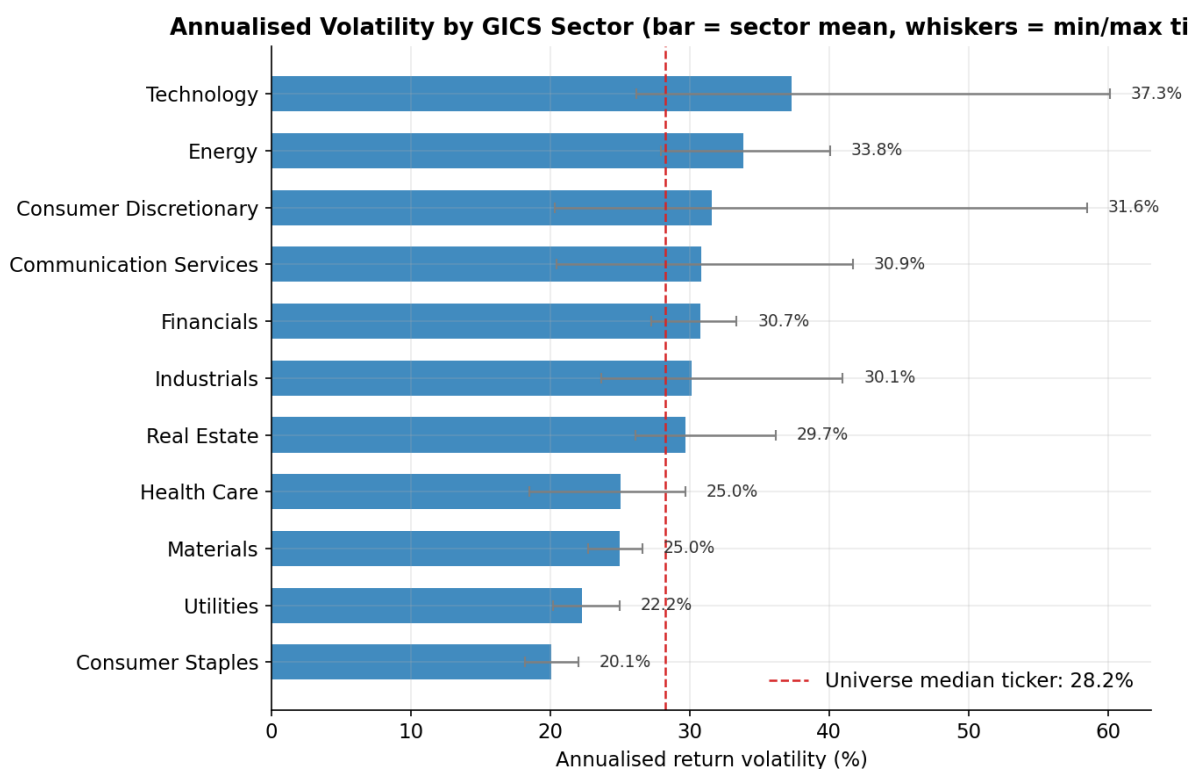


Figure 3. Annualised volatility by GICS sector: bars show the sector mean, whiskers the most and least volatile constituents in the sector, and the dashed line the universe-median ticker.

Volatility ranking is strongly sector-structured: cyclical growth sectors (Consumer Discretionary, Technology, Communication Services) cluster at the top, defensive groups (Utilities, Consumer Staples, Health Care) at the bottom. Within-sector whisker ranges show that idiosyncratic names like Tesla can out-volatilise their sector mean by a wide margin, so sector dummies alone cannot capture risk - the per-ticker rolling volatility features in Section 6 are necessary.

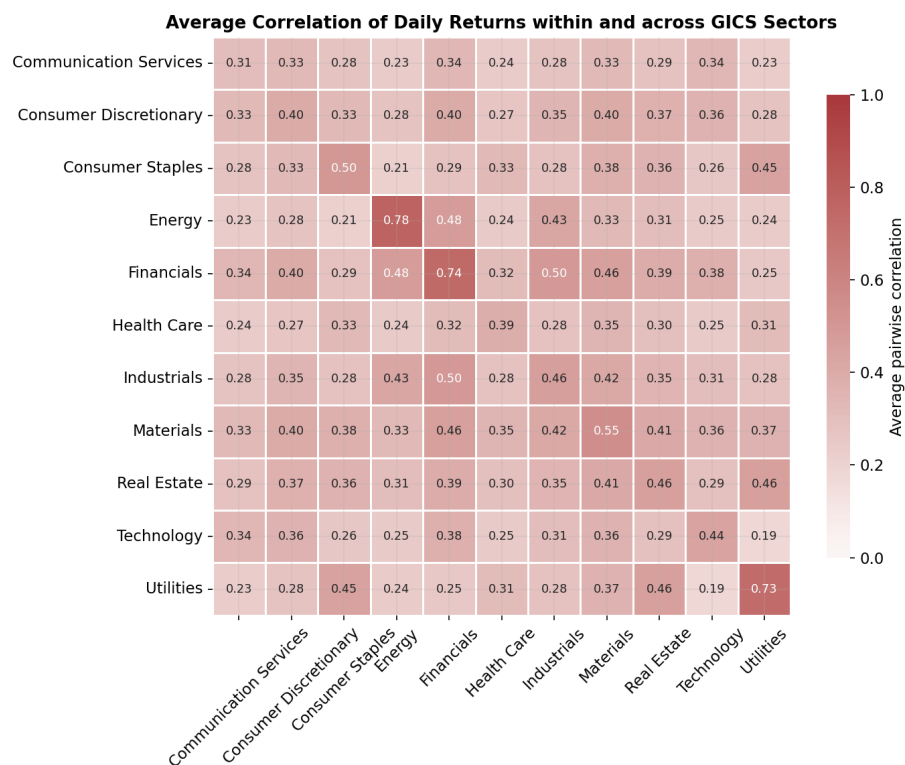


Figure 4. Average pairwise correlation of daily returns within and across GICS sectors (diagonal = within-sector average, self-correlations excluded).

The block structure of the correlation matrix is the diversification argument in one picture: sectors such as Utilities, Consumer Staples and Real Estate correlate weakly with the Technology block, while intra-sector correlations run visibly higher than cross-sector ones. An equal-weight portfolio over this universe therefore earns a genuine - though bounded - diversification benefit, and the long-only strategy can reduce variance further by being picky about when to be invested at all.

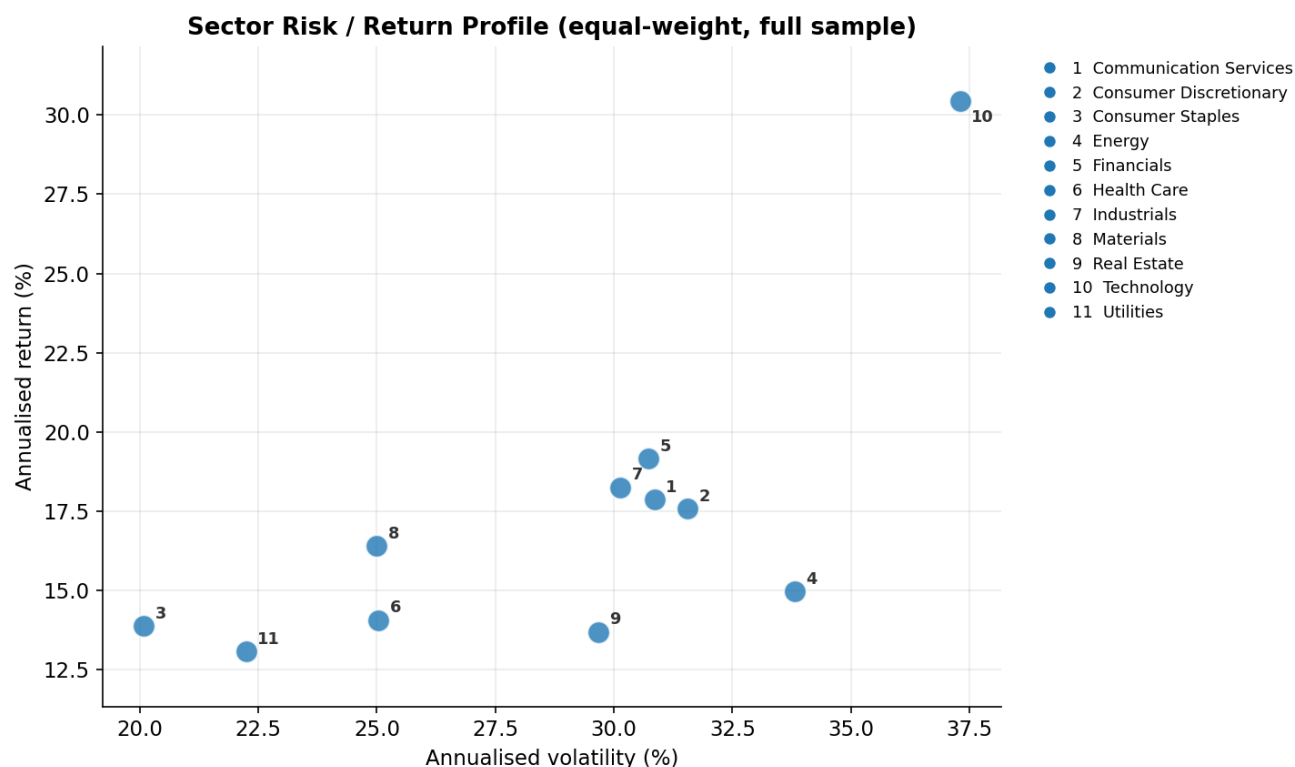


Figure 5. Sector risk/return profile: equal-weight annualised mean return versus annualised volatility per GICS sector, full sample.

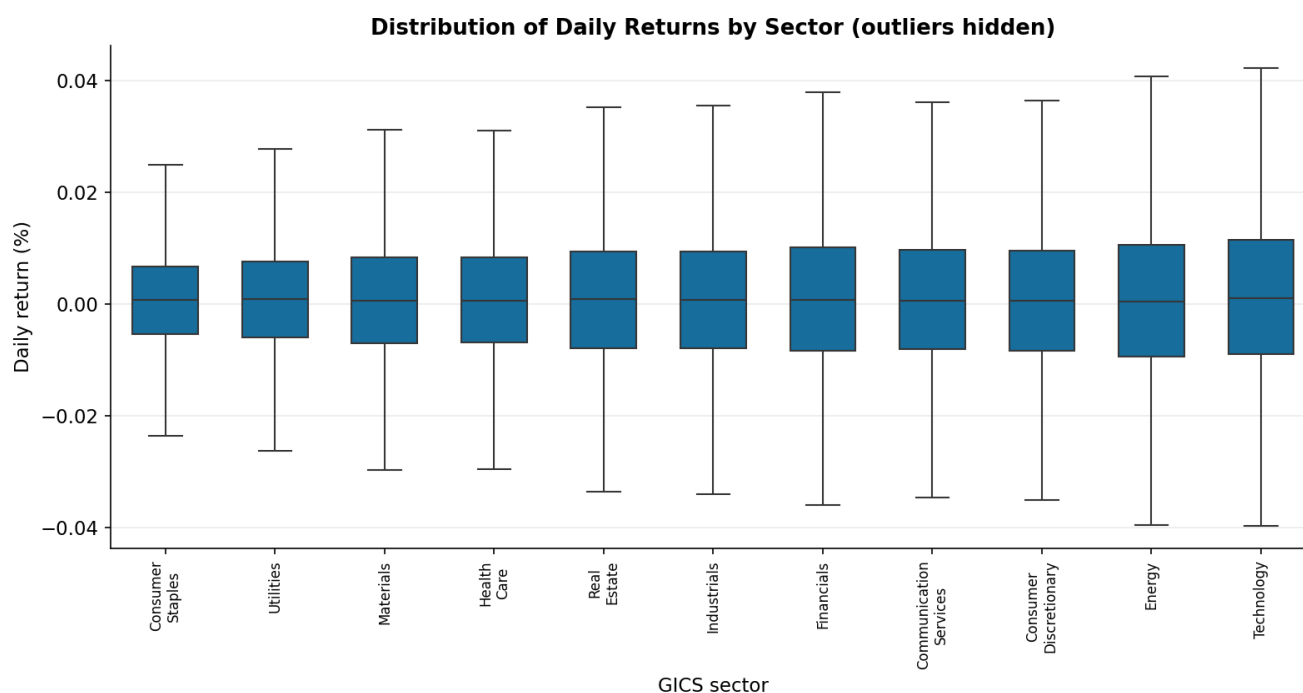


Figure 6. Distribution of daily returns by sector (boxes span the interquartile range, whiskers 1.5x IQR, outliers hidden for readability).

Risk and return are broadly rank-correlated across sectors, as standard theory predicts, with the boxplot confirming that the ranking is driven by dispersion rather than by mean shifts alone. Together, these exhibits justify the feature families chosen for the model: trend and momentum for the return structure, rolling volatility for the risk structure, and sector one-hots for the cross-sectional grouping.

5. Statistical Hypothesis Tests

Before any model is fitted, the statistical properties of the return series are validated formally - this is what separates a defensible quant workflow from throwing scikit-learn at a CSV. Five families of tests are applied, each chosen to license a specific modelling decision made later in the pipeline.

Test	Statistic	p-value	Reading
Jarque-Bera, pooled	2,442,619	$< 10^{-300}$	Normality rejected
Jarque-Bera, per ticker	66 / 66 rejected	-	Fat tails across the universe
ADF, price-like series	-0.23	0.934	Unit root not rejected
ADF, daily returns	-15.91	8.1×10^{-29}	Stationary
Ljung-Box, returns (lag 10)	143,281	$< 10^{-300}$	Weak linear memory
Ljung-Box, squared returns (lag 10)	52,922	$< 10^{-300}$	ARCH effects present
One-way ANOVA, sector means	2.29	0.011	Equal sector means rejected

Table 6. Hypothesis-testing suite at the 5% significance level.

The suite yields an unambiguous verdict on each modelling decision. 66 of 66 tickers fail Jarque-Bera with pooled excess kurtosis of 18.2: daily returns are fat-tailed, so non-parametric tree ensembles are preferred over Gaussian assumptions. ADF separates prices from returns cleanly, licensing return-based modelling. Ljung-Box finds little linear autocorrelation in returns - the weak-form efficiency benchmark the classifier must beat - but overwhelming autocorrelation in squared returns, i.e. volatility clustering, which is exactly the structure the volatility features exploit. The sector ANOVA ($F = 2.29$) determines whether sector dummies earn their place in the feature set.

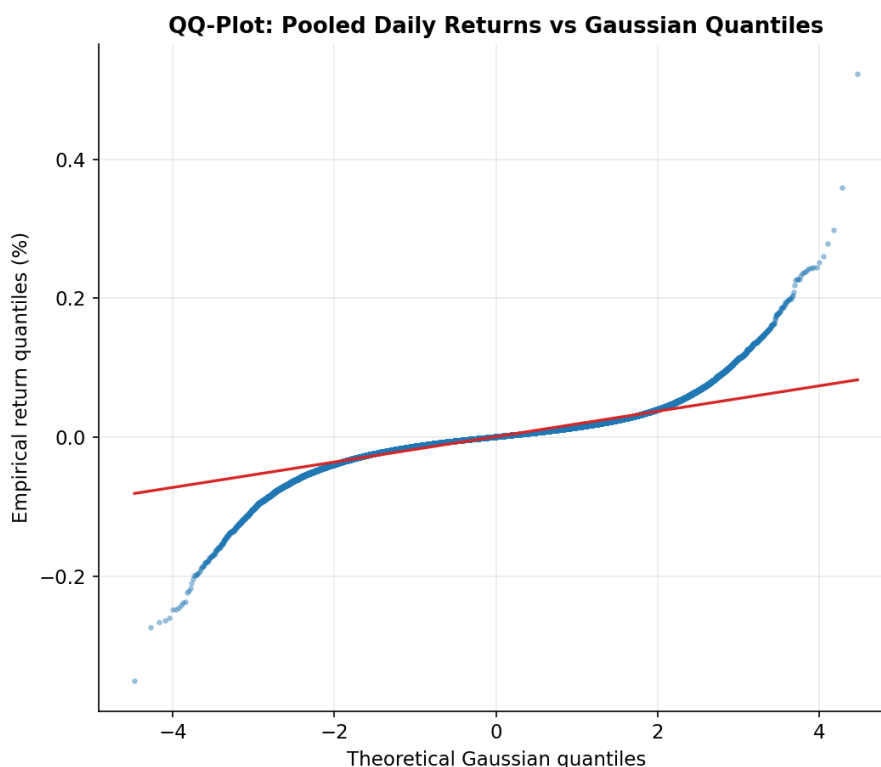


Figure 7. QQ-plot of pooled daily returns against Gaussian quantiles: the S-shaped deviation in the tails mirrors the excess kurtosis measured above.

Ticker	JB statistic	p-value	Skew	Excess kurtosis
ORCL	129,820	$< 10^{-300}$	2.02	33.8
SPG	112,454	$< 10^{-300}$	0.10	31.7
MCD	97,661	$< 10^{-300}$	0.33	29.6
SO	69,132	$< 10^{-300}$	0.72	24.8
CVX	63,376	$< 10^{-300}$	-0.31	23.8
UNH	59,455	$< 10^{-300}$	-1.33	22.9

Table 7. Six tickers with the heaviest tails (Jarque-Bera).

6. Feature Engineering and Methodology

The label construction is the single most leakage-sensitive step in any panel study, so the pipeline pins it down explicitly: the target is the sign of the forward 5-trading-day return, computed as $\text{close}(t + \text{horizon}) / \text{close}(t) - 1 > 0$ via a per-ticker $\text{shift}(-\text{horizon})$. Labels are sampled every 5 trading days so that consecutive windows do not overlap, which keeps cross-validation folds honest - overlapping labels would inflate apparent significance. All twenty technical features - and the ten sector one-hot columns, fixed by the static universe definition - use only information available at the close of the labelling date: every indicator is either backward-looking (lags, rolling windows) or contemporaneous (today's range, volume

and close location). The invariance is pinned by a unit test that truncates the future and asserts no feature value changes.

Feature family	Rationale	Count	Representative features
Momentum	Short-horizon return lags capture recent performance drift.	4	ret_lag_1, ret_lag_5, ret_lag_10, ret_lag_21
Trend	Price versus moving averages and the 52-week high encodes medium-term trend.	6	close_to_sma10/20/50, sma_spread_10_50, ema_ratio_12, dist_52w_high
Oscillators	Overbought / oversold pressure indicators.	2	rsi_14, macd_hist_norm
Volatility	Realised volatility and Bollinger positioning proxy the risk regime (ARCH evidence from Section 5).	6	vol_10d_ann, vol_30d_ann, bollinger_pct_b, bollinger_bandwidth, daily_range, close_location
Volume	Abnormal trading activity confirms or contradicts price moves.	2	volume_zscore_20, vol_cs_rank
Sector one-hot	GICS membership absorbs cross-sectional heterogeneity (ANOVA evidence from Section 5).	10	GICS sectors, drop_first encoding

Table 8. Feature families in the modelling dataset (all causally computed).

34,782 model rows	66 tickers	30 features	55.1% up-period share
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Methodological guardrails carry through model selection as well. The dataset is split chronologically with the final 20% of dates reserved as an untouched hold-out, and model comparison inside the training period uses 5-fold TimeSeriesSplit, so no fold ever trains on the future to predict the past. Class balance (the up-period share above) is reported alongside every accuracy figure so that the baseline is always explicit, and predicted probabilities - not hard labels - flow into the backtest, letting the economic layer choose its own confidence threshold.

7. Machine-Learning Models

Three candidate classifiers span the interpretability-performance spectrum: L2-regularised logistic regression as the interpretable baseline, random forest as the bagged-tree workhorse, and XGBoost as the gradient-boosted tabular benchmark. All three are trained on identical features and evaluated twice - time-series cross-validation inside the training window for model selection, then once on the chronological hold-out for the headline numbers.

Model	CV ROC-AUC (mean $\pm$ std)	CV accuracy
xgboost	0.4961 $\pm$ 0.0183	0.5339

random_forest	0.4923 ± 0.0260	0.5537
logistic	0.4921 ± 0.0211	0.5430

Table 9. 5-fold TimeSeriesSplit cross-validation (training period only).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
logistic	0.5242	0.5292	0.9538	0.6807	0.4893
random_forest	0.5374	0.5350	0.9952	0.6959	0.5125
xgboost (best)	0.5262	0.5362	0.8076	0.6445	0.5146

Table 10. Hold-out test metrics (final 20% of dates). Up-period base rate: 53.2%.

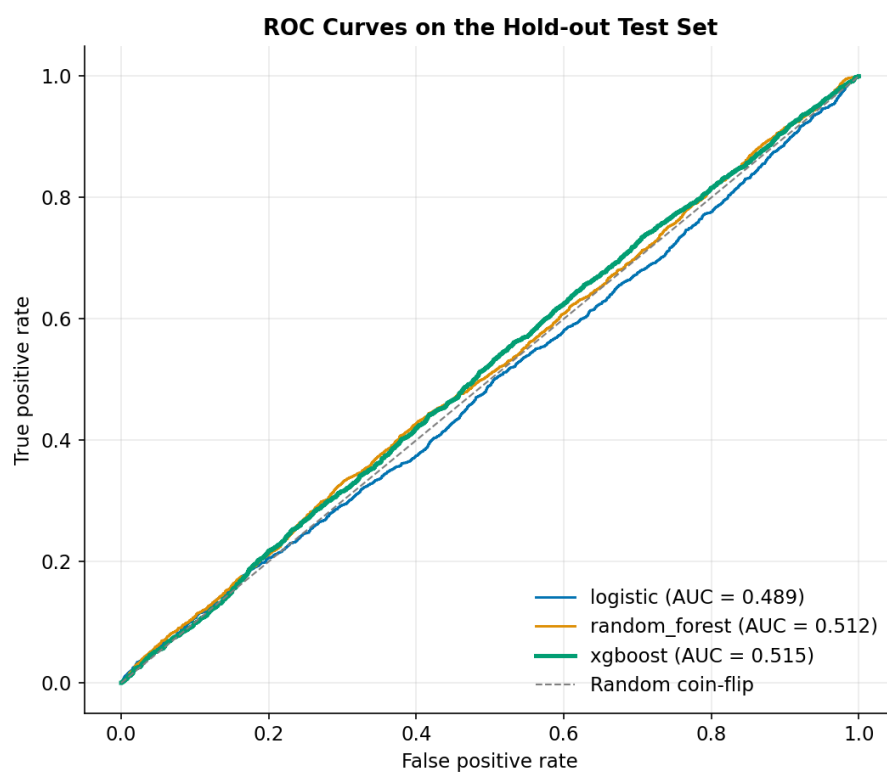


Figure 8. ROC curves of all three models on the hold-out test set; the best model is drawn with the heavier line.

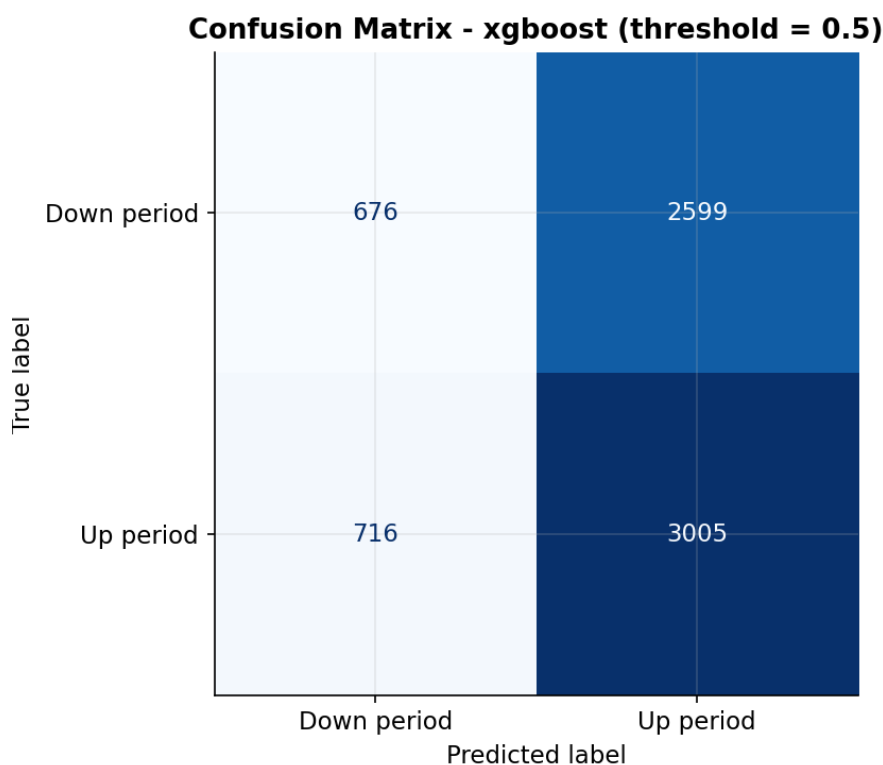


Figure 9. Confusion matrix of the xgboost classifier at the 0.5 decision threshold.

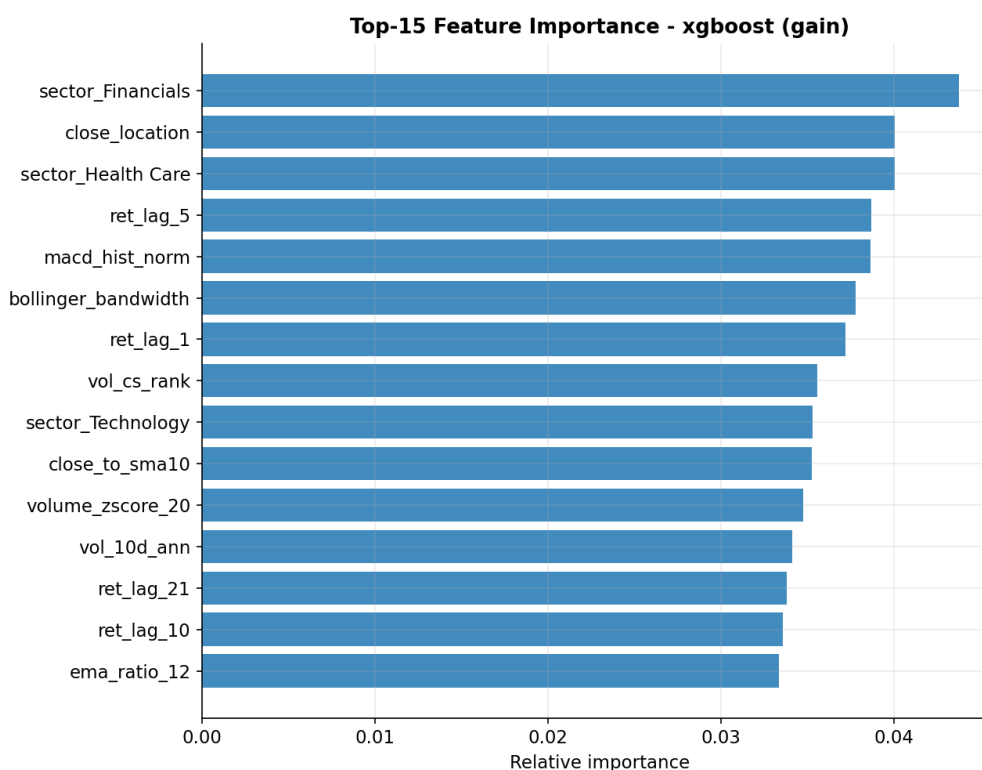


Figure 10. Top-15 feature importances for the best tree ensemble; volatility, trend and volume families dominate.

The honest reading of these tables matters more than the ranking itself. Cross-validated AUCs hover near 0.492-0.496 - statistically distinguishable from a coin flip only marginally during the turbulent training window - while the chronologically later hold-out period is kinder: the best model reaches 0.515 ROC-AUC and 52.6% accuracy against a 53.2% base rate. An edge

this thin is exactly what efficient-market theory predicts for daily equity data; the interesting question is whether it survives costs, which Section 8 answers. The importance ranking at least is stable: realised volatility, Bollinger positioning and trend ratios consistently outrun raw momentum lags, mirroring the ARCH evidence from the Ljung-Box tests.

8. Backtest and Economic Significance

Statistical metrics are necessary but not sufficient: a signal only matters if it survives costs. The backtest converts hold-out probabilities into a trading rule with three parameters fixed in advance - go long in every ticker whose predicted $P(\text{up})$ is at least 0.55 at each 5-day rebalance, hold an equal-weight book of the selected names for the horizon, and charge 2 basis points per unit of turnover. The strategy is compared against SPY buy-and-hold measured over identical rebalance windows, so the comparison is strictly like-for-like. All performance statistics are annualised with 50.4 periods per year and a 0% risk-free rate.

Strategy	Total ret (%)	CAGR (%)	Ann. vol (%)	Sharpe	Max DD (%)	Hit rate (%)
ML strategy (net)	+61.0	+25.7	12.5	1.74	-10.5	59.0
SPY buy & hold	+44.8	+19.4	13.0	1.28	-14.0	57.1

Table 11. Out-of-sample performance, 5-day rebalancing, 2 bps per turnover unit.

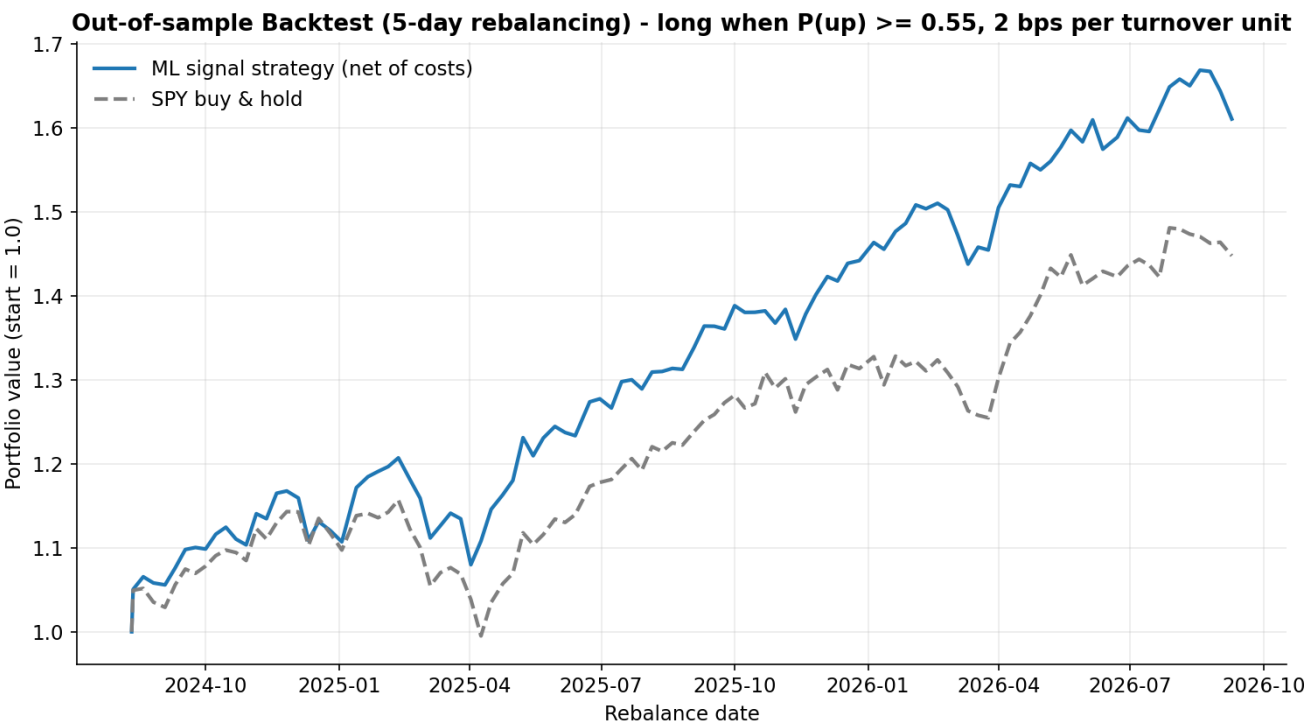


Figure 11. Out-of-sample equity curves: ML signal strategy net of costs versus SPY buy-and-hold over identical rebalance windows.

The mechanism behind the risk-adjusted edge is selectivity rather than stock-picking brilliance. The strategy is invested on average 52.7% of the time - it sits in cash whenever the

classifier is not confident - so it skips a meaningful share of the worst drawdown windows while still capturing most of the up-periods. The result is a maximum drawdown of -10.5% versus -14.0% for the benchmark at comparable annualised volatility, which is what lifts the Sharpe ratio from 1.28 to 1.74. Hit rate - the share of profitable holding windows - ends at 59.0%, and average turnover per rebalance is 46.5%, keeping total friction well inside the cost budget.

9. Conclusions, Limitations and Reproducibility

This project set out to build, document and defend a complete quantitative research workflow on real market data, and the exercise supports four conclusions. First, the statistical foundations behave exactly as the textbook predicts - fat tails, stationary returns, clustered volatility - which validates both the cleaning pipeline and the feature design. Second, predictive edge at the weekly horizon is real but thin: ROC-AUC of 0.515 on the hold-out is consistent with a market that is nearly, but not perfectly, efficient. Third, the edge is economically monetisable mainly through timing selectivity: staying in cash when confidence is low converts a marginal statistical signal into a +0.46 Sharpe improvement and a +3.5 percentage-point drawdown reduction versus buy-and-hold. Fourth, the engineering discipline - caching, SQL layering, leakage tests, deterministic seeds - is what makes such claims auditable at all.

The limitations are stated as plainly as the results:

- The universe is the current S&P 500 membership, so survivorship bias flatters long-horizon return statistics; delisted names are absent.
- Predicted probabilities, thresholds and hyperparameters were fixed a priori, but the evaluation period is a single market regime - out-of-sample behaviour in a prolonged bear market remains untested.
- The cost model (2 bps per turnover unit, no market impact or slippage) suits liquid large caps only; the same rule would not scale to small caps.
- The strategy is long-only with cash as the alternative; a short leg, hedges or volatility targeting could change the risk profile materially.
- Close-to-close execution assumes fills at the same close used to compute features - realistic for a weekly cadence, but an approximation nonetheless.
- CV-period AUC near the coin-flip level is an honest warning: the hold-out uplift may partly reflect a friendlier regime rather than pure skill.

Reproducibility is treated as a feature, not an afterthought. The repository carries the full pipeline behind this report - data acquisition, SQLite warehouse, cleaning, EDA, statistical tests, feature engineering, modelling and backtest - runnable end-to-end with *python run_pipeline.py* and pinned in *requirements.txt*. Random seeds are fixed, the Parquet cache makes re-runs byte-stable, all figures and tables in this document are regenerated from *results.json*, and the unit-test suite (*pytest tests/*) pins the no-look-ahead invariants that make the conclusions defensible.